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A strontium lattice clock with both stability and uncertainty below 5×10^{-18}

Jie Li^{1,2}, Xing-Yang Cui^{1,2,3}, Zhi-Peng Jia^{1,2}, De-Quan Kong^{1,2}, Hai-Wei Yu^{1,2}, Xian-Qing Zhu^{1,2}, Xiao-Yong Liu^{1,2}, De-Zhong Wang^{1,2}, Xiang Zhang^{1,2}, Xin-Yun Huang^{1,2}, Ming-Yi Zhu^{1,2}, Yu-Meng Yang^{1,2}, Yi Hu^{1,2}, Xiang-Pei Liu^{1,2}, Xiao-Min Zhai^{2,3}, Peng Liu^{2,3}, Xiao Jiang^{1,2,3} , Ping Xu^{1,2,3}, Han-Ning Dai^{1,2,3,*} , Yu-Ao Chen^{1,2,3,*}  and Jian-Wei Pan^{1,2,3,*} 

¹ Hefei National Research Center for Physical Sciences at the Microscale and School of Physical Sciences, University of Science and Technology of China, Hefei 230026, People's Republic of China

² Shanghai Research Center for Quantum Sciences and CAS Center for Excellence in Quantum Information and Quantum Physics, University of Science and Technology of China, Shanghai 201315, People's Republic of China

³ Hefei National Laboratory, University of Science and Technology of China, Hefei 230088, People's Republic of China

E-mail: daihan@ustc.edu.cn, yuaochen@ustc.edu.cn and pan@ustc.edu.cn

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Abstract

We report the realization of the closed-loop operation of an optical lattice clock based on ^{87}Sr atoms. A cavity-stabilized 698 nm laser is used to probe the $^1S_0 \rightarrow ^3P_0$ clock transition of strontium atoms trapped in optical lattices. Therein, we obtain a Fourier-limited Rabi spectrum with 0.6 Hz linewidth. The two transitions from $m_F = \pm 9/2$ ground states are alternatively interrogated to realize the closed-loop operation of the clock, and the clock laser light is frequency-stabilized to the center of the two transitions. Based on the interleaved measurement, the frequency instability of a single optical clock is optimized for the Dick effect, which is demonstrated to be $4.5 \times 10^{-16}/\sqrt{\tau}$, with τ being the averaging time for measurement. Further, we build another similar setup of the strontium lattice clock, which is used for the asynchronous comparison between the two clocks, where the stability is measured as 2.1×10^{-18} at 47 000 s. Moreover, we carefully calibrate the systematic effects of the Sr1 optical clock, and the total uncertainty is evaluated as 4.4×10^{-18} .

Supplementary material for this article is available [online](#)

Keywords: optical clocks, clock stability, clock uncertainty, spectroscopy

1. Introduction

In recent years, atomic clocks based on narrow optical transitions have made significant breakthroughs, demonstrating unprecedented accuracy and stability as frequency references. Currently, state-of-the-art optical clocks have achieved fractional frequency instabilities at several 10^{-17}

in 1 s averaging time [1] and systematic uncertainties of a few parts in 10^{-18} [2], surpassing their microwave counterparts for the definition of the international system of units (SI) second [3] by more than two orders of magnitude, which opens up the possibility of a redefinition of the SI second in the future [4]. The increasing precision of optical clocks allows height differences of a millimeter to be measured to test the gravitational redshift [5], enabling applications in the field of relativistic geodesy [6–8]. Moreover, dedicated optical links and networks for connecting distant optical clocks are being

* Authors to whom any correspondence should be addressed.

established [9], facilitating new tests of fundamental physics, such as detecting gravitational waves [10] and searching for dark matter [11, 12].

To date, there are mainly two different types of optical clocks being developed worldwide, i.e. ion clocks based on single ion [13, 14] and lattice clocks based on neutral atoms [15–17]. Compared with ion clocks, lattice clocks have much lower quantum projection noise (QPN) [18] for interrogating many atoms [19], requiring much shorter measurement times. Lattice clocks based on neutral atoms with a spin- $\frac{1}{2}$ system, e.g. ^{87}Sr [2, 15, 17] and ^{171}Yb [7, 20], have already been developed. In the future, employing long-distance fiber links [21, 22], optical clock networks will be established. Among various options, the strontium clock [23, 24] is recognized as one of the most promising systems for achieving high performance with reliable laser schemes for cooling, manipulating, interrogating, and detecting. Its clock transition $^1S_0 \rightarrow ^3P_0$ at 429 THz and linewidth ~ 1 mHz was endorsed by the International Committee for Weights and Measures as a secondary representation of the second [4].

In view of these advances, the central objective of an optical clock is to suppress the low-frequency laser noise by locking on an atomic reference. Thus, its stability performance is limited by a balance of QPN and other noise processes, such as laser phase noise [25] and the Dick effect [26, 27]. The laser phase noise is fundamentally limited by the thermal noise floor [28], as well as by other technical noises from the environment [29]. The Dick effect is due to the dead time in clock interrogation sequences for preparing atomic samples [27].

In this work, we report the realization of a lattice clock based on ^{87}Sr in which atoms are laser-cooled and confined in a one-dimensional (1D) optical lattice. A clock laser at 698 nm, frequency-stabilized to a 30 cm long ultralow-expansion glass (ULE) cavity, is used to interrogate the atoms. We obtain high-contrast Fourier-limited Rabi spectra with sub-Hz linewidths. By alternatively interrogating the two transitions from $m_F = \pm 9/2$ ground states, feedback to the clock frequency is applied with low-noise servos. Hence, the clock frequency is stabilized to the atomic clock transition. To estimate the closed-loop performance of our lattice clock, we implement the interleaved measurement. By optimizing the clock sequence according to the frequency noise in our clock laser, we achieve a stability of $4.5 \times 10^{-16}/\sqrt{\tau}$, approaching the limited performance of the Dick effect. To accurately evaluate the systematic stability of a single clock, we perform an anti-synchronous comparison between two independent clocks, demonstrating the stability of 2.1×10^{-18} in 47 000 s. In addition, we evaluate the systematic uncertainty of 4.4×10^{-18} .

2. Experimental method

The experimental setup for the strontium lattice clock is sketched in figure 1. It consists of a compact atomic beam source [30], a science chamber, a set of cooling and manipulation lasers, and a high-performance clock laser reference cavity. The strontium atoms are first heated into hot

atomic vapor in the oven at a temperature of 800 K, and ejected through a nozzle composed of capillary array with a divergence half-angle of 17 mrad. Then, the atoms with velocities lower than 260 m s^{-1} are decelerated to below 40 m s^{-1} by passing through a permanent magnet Zeeman slower, composed of an array of small permanent magnets. At the end of the Zeeman slower, the divergence angle of the atomic beam is further squeezed by a two-dimensional (2D) transverse cooling light, which is perpendicular to the direction of atomic motion. Afterward, the atoms with low velocities are deflected by 20° with a 2D MOT deflector and enter the science chamber through a differential tube. The ^{87}Sr atomic flux sent into the science chamber is estimated as $1.4 \times 10^8 \text{ atoms s}^{-1}$. We note that the deflector plays a role as a laser-controlled fast switch for the atomic beam flux, preventing unwanted high-pressure hot atoms and direct blackbody radiation (BBR) by the oven from entering the science chamber.

The science chamber is an 8 inch extended spherical octagon chamber (Kimball Physics Inc.), working at 1.1×10^{-10} mbar. The chamber has multiple optical access for laser cooling and manipulation with anti-reflection-coated viewports, whose temperatures are actively controlled by thermoelectric coolers and water-cooled bases, forming a temperature-stabilized BBR environment. A pair of water-cooled anti-Helmholtz magnetic coils (in the y - z plane) is installed above the re-entrant viewports for magnetic-optical trapping (MOT). In addition, with the background magnetic field being compensated by three pairs of rectangular coils, the bias field for clock interrogation is provided by another pair of Helmholtz coils.

The lasers for cooling and trapping the strontium atoms are delivered from commercial laser systems (Toptica Inc.). The 461 nm laser for the blue MOT is derived from a second-harmonic-generation (SHG) of high power 922 nm diode laser (TA-SHG Pro). The repumping lasers (679 nm and 707 nm) and the red MOT lasers (689 nm) are generated by diode lasers directly (DL Pro). The 813 nm laser for optical lattices is provided by a sum frequency generation laser, free from the additional light shift caused by the amplified spontaneous emission (ASE), and a volume Bragg grating (bandwidth < 35 pm, OptiGrate Inc.) is also employed to further suppress the potential residual background spectrum [31, 32]. All of the above lasers are stabilized to a 10 cm optical cavity (Stable Laser Systems Inc.) using a developed scheme of multiwavelength frequency stabilization [33].

In the science chamber, the atomic cloud is first trapped near the zero point of the 50 G cm^{-1} gradient magnetic field by the 461 nm trapping beam and 679 nm and 707 nm repumping beams. To achieve a complete repumping cycle, a fiber electro-optical modulator is used to modulate the frequency of the light [34] instead of fast sweeping of the laser piezo or complex optics modulated by acoustic-optic modulators (AOMs). This repumping scheme is more miniaturized and robust for trapping atoms. After holding the blue MOT for 600 ms, the temperature of the atomic cloud is reduced to 2 mK. Then, atoms are further cooled to a temperature of $2 \mu\text{K} \sim 3 \mu\text{K}$ in the red MOT from broadband to single-frequency

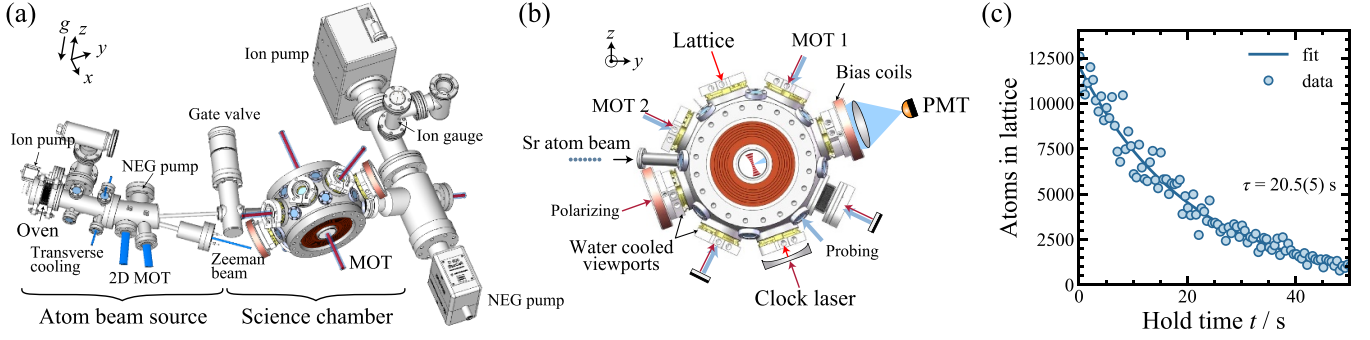


Figure 1. Atomic physics system and atomic cooling. (a) Structure diagram of atomic physics system which consists of atomic source and science chamber. Atomic source contains atomic oven, Zeeman slower, transverse cooling, 2D MOTs, hot window and differential tube. The direction of gravity is along the $-z$ -axis. (b) Optics for atom cooling in science chamber. Three MOT beams including 461 nm and 689 nm light are orthogonal to each other. Polarizing light for state preparation is incident along the direction of the bias magnetic field. The retro-reflected lattice light is injected at an angle of 22.5° to the direction of gravity. And 698 nm the clock laser coincides with the lattice light. (c) The lifetime of one-dimensional lattice. The atoms number in lattice is about 12 000, and the lifetime is about 20.5 s by a single exponential fitting.

cooling by a 689 nm trapping beam and stirring beam for 250 ms [34]. Sequentially, the cold atom cloud is loaded into an 813 nm 1D lattice with a diameter of $70 \mu\text{m}$. As shown in figure 1(c), 12 000 atoms can be loaded into the 1D optical lattice, and the number of atoms can be controlled by the on-time of the deflector in the blue MOT stage. The lifetime of the atoms in the lattice is estimated to be ~ 20.5 s, which is consistent with the background pressure of 1.1×10^{-10} mbar.

3. Sub-Hz clock spectroscopy

The clock laser system is shown in figures 2(a) and (b). There are four functional modules in the entire optical system. In the first module, an ultrastable 30 cm-long cavity (U1) with a finesse of $\sim 320\,000$ is used as a frequency reference, composed of an ultralow-expansion glass spacer and fused silica mirrors. The cavity has an expected thermal noise-limited stability [28] of $\sim 1 \times 10^{-16}$ at 1 s. The frequency of the clock laser is stabilized to the cavity by the Pound–Drever–Hall locking method, yielding a frequency instability of 2.6×10^{-16} at a 1 s averaging time. The instability is mainly limited by the vibration noise and evaluated by a ‘three-cornered hat’ method for the frequency beat with the other two ultrastable lasers. Furthermore, AOM 1 on the frequency-locked loop is used to compensate for the long-term linear drift of the cavity. The frequency of laser output ν_{laser} can be written as $\nu_{\text{laser}} = \nu_{\text{cavity}} - 2\nu_{\text{AOM1}}$, where ν_{cavity} is the frequency of cavity mode and ν_{AOM1} is the frequency of RF signals. To suppress the drift of ν_{cavity} , the temperature of the cavity has been set to the zero-thermal-expansion temperature. The RF sources of AOM 1, 2, and 4 are supported by homemade high-precision direct digital frequency synthesis devices (DDSs), of which the resolution and noise are on the order of millihertz. By compensate the linear cavity drift to ν_{AOM1} , the residual cavity drift can be controlled within 1 mHz s^{-1} .

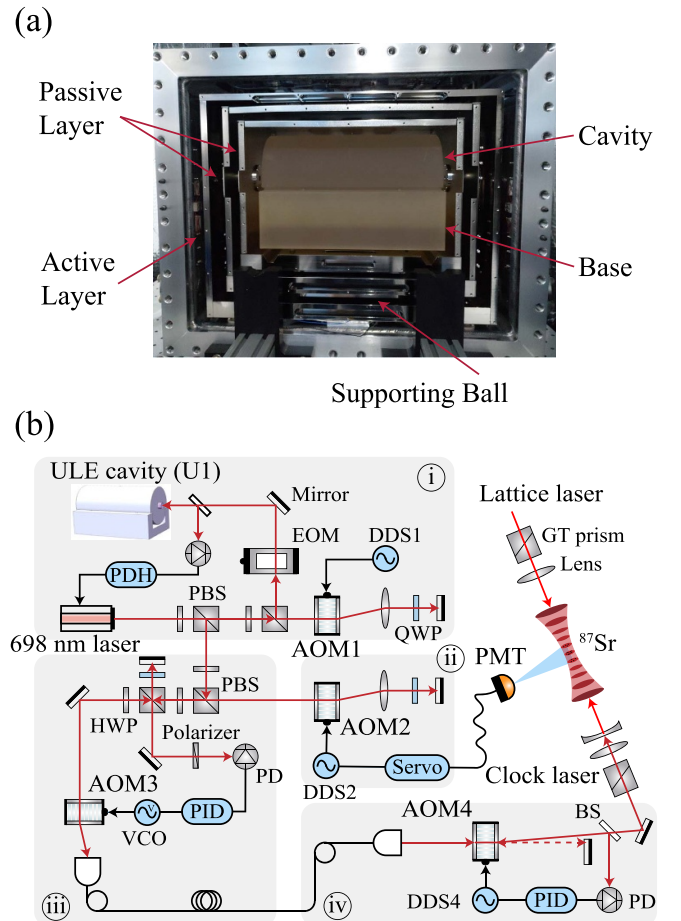


Figure 2. (a) Structure of the vacuum chamber with temperature stability layers of the U1 cavity. (b) Schematic diagram of frequency locking and atomic interrogation of 698 nm ultrastable lasers. Four functional modules are implemented in the optical path: (i) PDH frequency locking and frequency compensation for linear drift, (ii) frequency closed-loop feedback, (iii) fiber phase noise and AOM phase chirp cancellation, (iv) clock interrogation and power stabilization.

The second module is used to achieve a closed loop of the frequency feedback of the optical clock, in which a proportional-integral-derivative (PID) servo feeds back to the RF frequency of AOM 2. According to the excitation fraction of atoms, the clock beam frequency is locked to the atomic clock transition.

Through the third module, the frequency-locked clock beam is transferred to the atomic table through a single-mode polarization-maintaining fiber and diffracted to the science chamber by AOM 4. However, phase noise will be introduced during the fiber transmission process [35], and a phase chirp [36] will be introduced during AOM switching. Phase noise cancellation is designed, in which the zero-order beam is reflected and transmitted in the fiber again, of which a fast photodiode detects phase noise. Any phase change in fiber transmission or AOM phase chirp can eventually be compensated by AOM 3 driven by a voltage-controlled oscillator scontrolled by a PID servo.

The fourth module is used for beam switching and power stabilization with AOM 4. Moreover, the lattice and clock beams are well aligned and co-polarized using two corresponding Glan–Taylor prisms parallel to the bias magnetic field axis. The population between the ground state and excited state of atoms in the lattice will be changed during the interrogation stage of the 698 nm clock, which can be obtained by the fluorescence signals collected by a photomultiplier tube (PMT) during atomic detection.

To achieve a closed loop, we should first search for the corresponding clock transition frequency. Using lattice sideband spectroscopy is an efficient method. The motion of atoms in an optical lattice can be described as a quantized harmonic oscillator, which brings out the vibration energy levels $|n_x, n_y, n_z\rangle$. The transition corresponding to $|n_z\rangle \rightarrow |n_z + 1\rangle$ is called the blue sideband, while the other transition corresponding to $|n_z\rangle \rightarrow |n_z - 1\rangle$ is the red sideband. Figure 3(a) shows the sideband spectra of atoms in an optical lattice, from which the temperature of the atom cloud is evaluated as 2.8 μ K and the depth of trap is 124 E_r according to theoretical fit [37].

The coherence of the clock laser limits the interrogation time. Moreover, a longer clock pulse means a narrower linewidth of carrier spectra using lower power of the clock beam. However, the presence of the background magnetic field will broaden the spectral lineshape and limit the coherence time, thereby reducing the frequency discrimination coefficient and affecting the operating performance of the optical clock. In our system, the residual background magnetic field is well canceled, which is less than 0.1% relative to the bias magnetic field [34].

During actual operation, it is difficult to measure a narrow linewidth spectrum because of the frequency noise and drift of the clock laser. Therefore, a process similar to the interleaved measurement is adopted, during which two cycles are used to lock the clock frequency, and another two cycles are used for spectra. As shown in figure 3(b), by keeping the π clock pulse T_p at 724.6 ms with a Rabi frequency of $2\pi \times 0.69$ Hz, the carrier spectra are measured with a full width at half maximum (FWHM) of approximately 1.1 Hz, which is consistent with the theoretical Fourier linewidth limit $0.8/T_p$. By decreasing

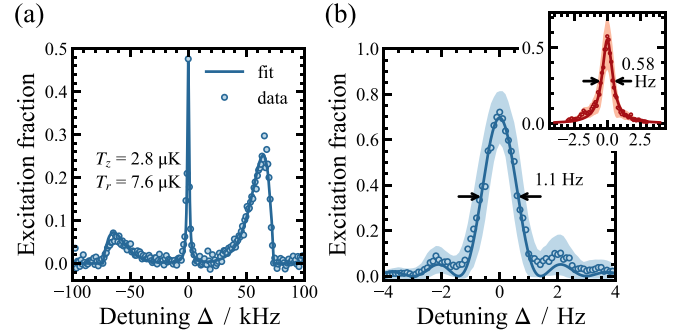


Figure 3. (a) Sideband spectra of atoms in the optical lattice. The blue curve is a fit of sideband line-shape [37], which evaluates the atom temperature as 2.8 μ K. (b) Clock carrier spectra with a 1.1 Hz linewidth using clock pulse of 724.6 ms and a 0.58 Hz linewidth using clock pulse of 1390 ms (inset). Every point is tested repeatedly for 100 times. The circle represents the mean value of the data and the shaded area represents the standard deviation of the data.

the clock beam's power further to achieve a longer π clock pulse T_p of 1390 ms, the carrier spectra are obtained with an FWHM of ~ 0.58 Hz, which is also consistent with the theoretical Fourier linewidth limit. Therefore, we have obtained the carrier spectra of the sub-Hz linewidth and ensured that the clock interrogation time is shorter than the Rabi oscillation decoherence time, which leads to the possibility of optimizing the stability of our optical lattice clock.

4. Analysis of clock laser noise

The Dick effect dominates the stability of our optical clock where the QPN is negligible [34]. For a clock interrogation time T_p in a clock cycle time T_c , the stability is limited by the dead time $T_d = T_c - T_p$ for atom cooling or detection in the operation of an optical lattice clock, which is called the Dick effect [27, 38]. It can be calculated as

$$\sigma_{\text{Dick}}^2 = \frac{1}{\tau} \sum_{k=1}^{\infty} \left| \frac{G(k/T_c)}{G(0)} \right|^2 S_y(k/T_c) \quad (1)$$

where S_y is the one-sided power spectral density of fractional frequency fluctuations of the local oscillator, and $G(f)$ is the standard Fourier transform of the sensitivity function [34].

To evaluate the Dick-effect-limited instability of our lattice clock based on equation (1), we obtain S_y from the cross-spectral density (CSD) [39]. The CSD measurement uses U1 laser to beat with two other independent ultrastable lasers (U2 and U3), yielding beat signals f_{12} and f_{13} , as shown in figure 4(a). Figure 4(b) shows the frequency fluctuations in the time domain of the beat signals f_{12} and f_{13} with a sampling rate of 20 Hz. Since the drifts of both U2 and U3 are not pre-compensated, the linear drifts of f_{12} , f_{13} , and f_{23} need to be deducted during postprocessing.

Figure 4(c) shows the CSD of U1 laser, which is obtained from f_{12} and f_{13} . The CSD has a prominent peak near 2.3 Hz and some other narrow peaks, with Lorentz fit $A_i \{1 + [(f - f_i)/(\Gamma_i/2)]^2\}^{-1}$, where A_i is the amplitude, f_i is the center

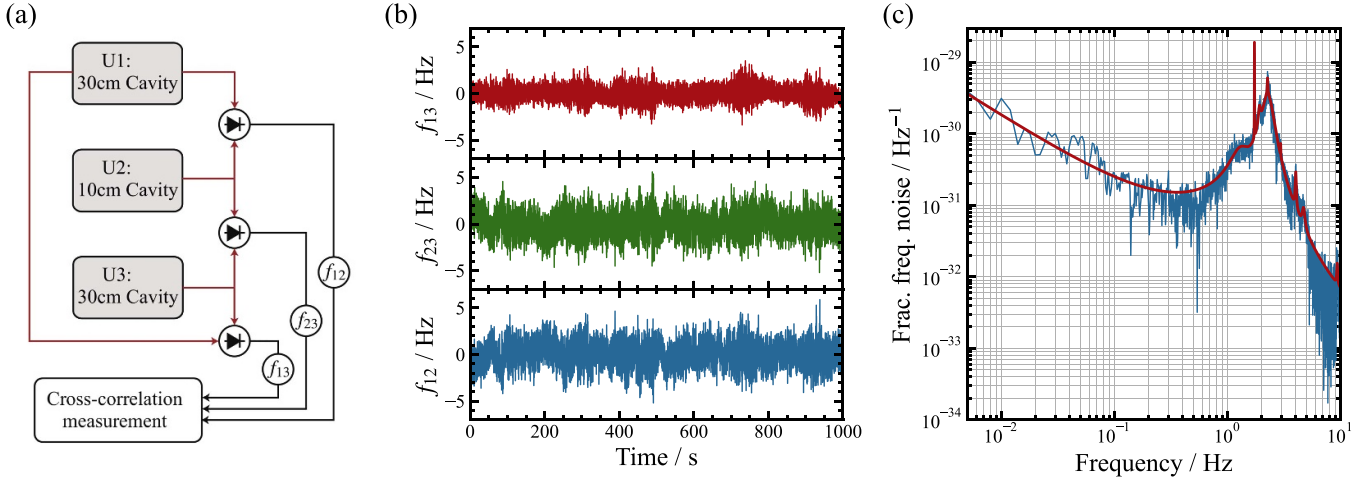


Figure 4. (a) Schematic of the cross-correlation measurement. (b) Beat signals between lasers correspondingly locked to three ultra-stable cavities: U1 (30 cm), U2 (10 cm) and U3 (30 cm). Linear drift and offset have been deducted in f_{13} , f_{23} , f_{12} . (c) Fractional frequency noise of U1 laser described by the CSD (blue line) from data in (b). There are nine peak fittings (red line).

frequency, and Γ_i is the linewidth of the i th peak. The two broadest peaks, determining the overall shape of CSD, are near 1.3 Hz with a linewidth of 0.57 Hz and 2.3 Hz with a linewidth of 0.54 Hz, and the highest and narrowest peak is near 1.75 Hz with a linewidth of 2.5 mHz. The fitting results here differ from the [1] because of the difference in frequency noise peaks related to the local geological and traffic environment. The presence of noise peaks degrades the stability of the optical lattice clock. We can adequately select the parameters of T_p and T_d to obtain a proper transfer function $G(f)$ to avoid the effects of noise peaks to optimize the stability.

5. Optimization of the Dick effect

During the closed-loop operation of the optical clock, one clock loop has four clock interrogations to eliminate the first-order Zeeman effect on stability caused by the drift of the background magnetic field. At the i th clock loop, the first two interrogations are used to lock the clock laser to ν_i^- of the $m_F = -9/2$ state, while the last two interrogations are used to lock the clock laser to ν_i^+ of the $m_F = +9/2$ state. The output ν_i of the clock is $(\nu_i^+ + \nu_i^-)/2$. However, estimating the instability of ν_i requires a frequency comparison between two clocks.

As shown in figure 5(a), to experimentally estimate the instability of our lattice clock, we perform an interleaved measurement for self-comparison [40, 41], in which two independent PIDs run alternately in time, steering the clock laser onto resonance with the atomic transition. Similarly, one comparison includes four interrogation cycles: PID-a runs to lock the frequency to ν^- in the first two cycles, and PID-b runs in the last two cycles to lock the clock frequency to ν^+ . The instability of the clock frequency can be estimated by Allan's deviation from a dataset of $(\nu_i^+ - \nu_i^-)/2$, denoted as $\{\nu_i^*\}$. In contrast, ν_i^+ and ν_i^- correspond to the same state in self-comparison, so the first-order Zeeman effect on ν_i^* can be

equivalently eliminated. If the frequency noise is white noise, the Allan deviation will decrease with the averaging time in the $\tau^{-1/2}$ trend.

To understand the relationship between the Dick-effect-limited instability and the dead time and spectroscopy time, we perform a theoretical calculation of the interleaved measurement scheme based on equation (1), as shown in figure 5(b). The longer the dead time at a fixed spectroscopy time, the worse the clock instability. More spectroscopy time will benefit stability for a fixed dead time. Interestingly, the existence of CSD noise peaks leads to the appearance of wrinkles on the contour plot. However, it is efficient to avoid the destructive effect of frequency noise by using a longer spectroscopy time. This indicates that better clock stability can be achieved with a longer pulse and shorter dead time.

Figure 5(c) shows the tendency of Allan deviation with averaging time while the optical lattice clock is running with a clock pulse of 450 ms and cycle time of 1520 ms, which indicates instability as $1.6 \times 10^{-15}/\sqrt{\tau}$. After an averaging time of $\sim 25\,000$ s, the self-comparison stability enters the 10^{-18} level. The experimental stability is larger than the theoretical stability because the experimental condition is near the noise peak. The stability is much larger than the thermal noise-limited stability, which indicates the possibility of better optimization.

The optimization has been executed from case (i) to case (iv). From case (i) to case (iii), the Dick-effect-limited stability is expected to improve significantly with a fixed dead time and increased spectroscopy time based on the theoretical calculation. Furthermore, the clock stability is evaluated by the interleaved measurement, which is shown in figures 5(c)–(e). Based on the results of the fits (blue lines) with the white frequency noise model, clock stability has been improved from $1.6 \times 10^{-15}/\sqrt{\tau}$ to $5.8 \times 10^{-16}/\sqrt{\tau}$ by increasing the π clock pulse to 1390 ms and holding the dead time at ~ 1080 ms. There is also a slight improvement in clock stability from case (iii) to case (iv). As shown in figure 5(f), clock stability has been further improved to $4.5 \times 10^{-16}/\sqrt{\tau}$ by decreasing the

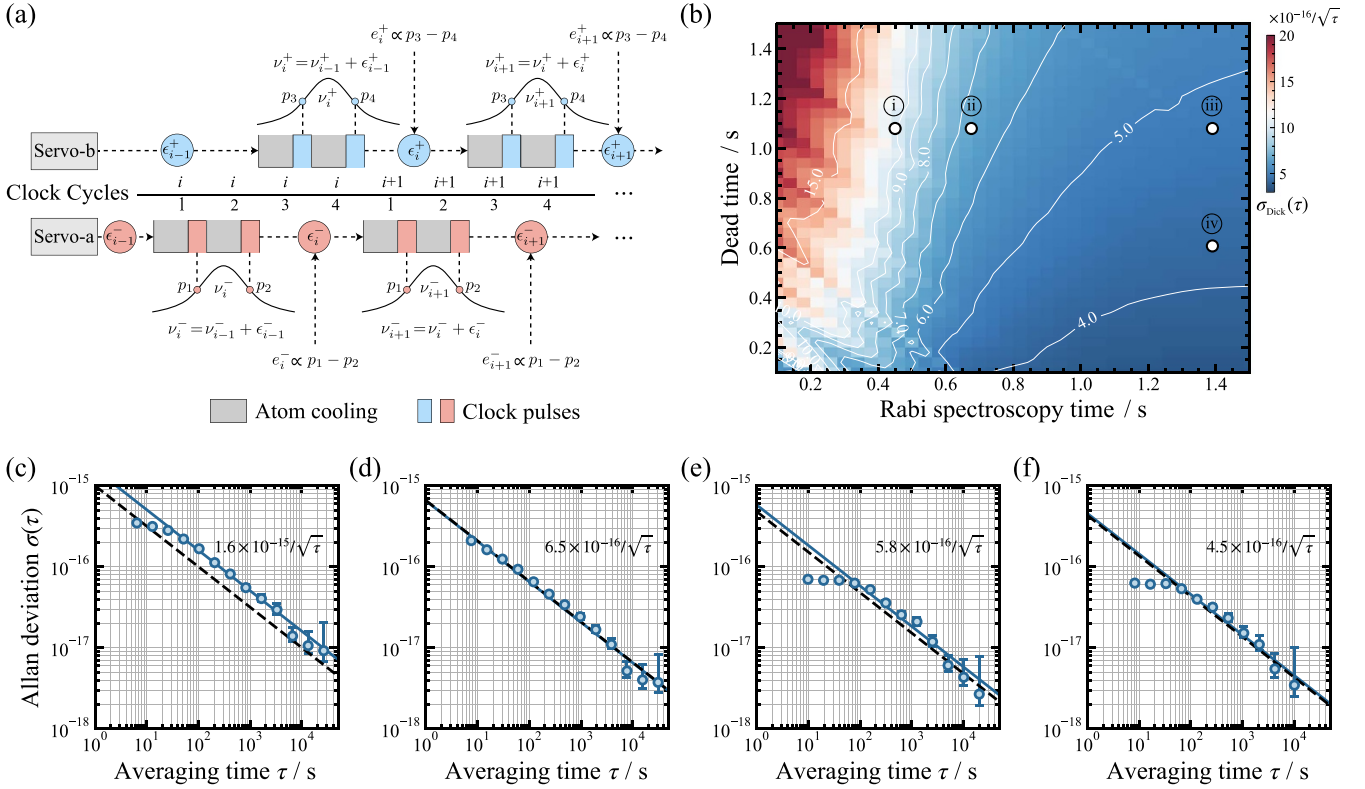


Figure 5. (a) Timing sequence of the interleaved measurement to evaluate clock stability in case of one optical clock. (b) Calculation of the Dick-effect-limited instability of an optical clock. The symbols (i)–(iv) represent four cases of different clock pulse and cycle time. Case (i): $T_p = 450$ ms, $T_c = 1520$ ms; case (ii): $T_p = 675.7$ ms, $T_c = 1760$ ms; case (iii): $T_p = 1390$ ms, $T_c = 2474$ ms; case (iv): $T_p = 1390$ ms, $T_c = 2074$ ms. (c)–(f) describe the instability of the optical lattice clock for cases (i)–(iv), respectively. Error bars represent the Allan deviation of the experimental data. The blue solid lines represent the fits by a white frequency noise model. The black dotted lines correspond to the theoretical Dick-effect-limited instability.

dead time to 684 ms. If the dead time is further reduced, the stability will still be slightly improved, but the cost is that the number of atoms will be decreased, which may introduce more detection noise. The experimental results agree with the theoretical limit, which verifies the correctness and validity of the theoretical calculation based on the CSD of the U1 laser.

6. Clock stability

Since the self-comparison cannot reflect the long-term drift of the clock, we use a frequency comparison between two similar optical clocks to accurately evaluate the performance of each clock. Therefore, we developed two independent atomic physics systems, both of which are 1D optical lattices, named Sr1 and Sr2. Figure 6(a) shows that the two clocks share the ultrastable laser locked on the U1 cavity, each with independent fiber noise cancellation. The physical systems of Sr1 and Sr2 are built in two laboratories with independent ambient temperatures, magnetic fields, and laser systems for atom cooling. Especially, two different lattice lasers filtered by volume Bragg gratings are used for atom trapping in the two systems, respectively, where the residual light shifts due to the ASE background spectra are independent. We estimate the systematic instability of a single optical clock using an anti-synchronous comparison method where clock spectroscopy

for the two systems has no temporal overlap, as shown in figure 6(b). Therefore, it is sensitive to all sources of instability, including the Dick effect, QPN, and other physical noise. The transistor-transistor logic (TTL) trigger signal is transmitted over a 70 m cable with a delay of less than 360 ns.

Assuming that the two clocks have identical noise levels but are uncorrelated, the single-clock instability can be inferred by analyzing $[\nu_1(t) - \nu_2(t)]/\sqrt{2}$, where $\nu_1(t)$ is the frequency record for the Sr1 clock and $\nu_2(t)$ is the frequency record for the Sr2 clock [1, 16]. In our setup, the ratio of a clock pulse to dead time is 1:1. A single clock instability is evaluated as $5.4 \times 10^{-16}/\sqrt{\tau}$ from figure 6(c). We note that long drifts in ambient temperature and magnetic fields limit the long-term stability of our clocks. Therefore, we calculate the BBR frequency shift correction and the second-order Zeeman shift correction by monitoring the temperature of the main chambers and the ambient magnetic field. After frequency shift correction of the data points, the anti-synchronous comparison demonstrates the stability of 2.1×10^{-18} in 47 000 s.

7. Clock uncertainty

Accurate determination of the unperturbed ^{87}Sr clock transition frequency requires deduction of energy shifts between the 1S_0 ground state and the 3P_0 metastable excited state by all

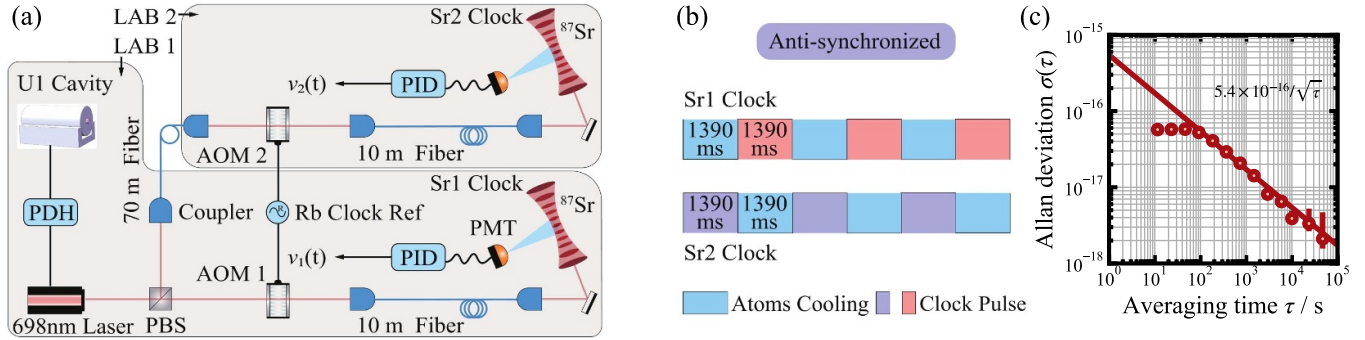


Figure 6. Frequency comparison between Sr1 and Sr2 clocks. (a) Experimental layout. The 698 nm laser is stabilized to a 30 cm ULE cavity using the Pound–Drever–Hall (PDH) technique. The clock light is sent to the Sr1 and Sr2 clocks over optical links with active path length stabilization. Two independent digital servos steer the laser frequency using AOMs to lock both probe paths to the corresponding atomic transition. The physical systems are built in two independent laboratories, named LAB 1 and LAB 2. (b) Anti-synchronous comparison. The two clock pulses are anti-synchronized to prevent common mode rejection of the Dick effect. (c) The total Allan deviation of $[\nu_1(t) - \nu_2(t)]/\sqrt{2}$, indicating a single clock instability of $5.4 \times 10^{-16}/\sqrt{\tau}$. Error bars indicate 1σ confidence intervals.

Table 1. The uncertainty of the Sr1 clock.

Systematic effects	Shift/ 10^{-18}	Uncertainty/ 10^{-18}
BBR	−4926.1	3.2
1st order Zeeman	0	0.8
2nd order Zeeman	−178.5	0.2
Density	−36.4	0.9
Lattice AC Stark	0.4	2.8
DC Stark	0	<0.1
Background gas	−1.5	0.2
Servo error	0	<0.1
Probe AC Stark	0	<0.1
Line pulling	0	<0.1
Lattice tunneling	0	<0.1
AOM phase chirp	0	<0.1
Total	−5142.1	4.4

systematic effects. Based on studies from the [2, 42], we consider the systematic effects listed in table 1 and evaluate the systematic uncertainty of the Sr1 clock.

7.1. BBR shift

The BBR shift is the largest systematic shift and a dominant source of uncertainty, which can be expressed as:

$$\Delta\nu_{\text{BBR}}(T) = \nu_{\text{sta}} \left(\frac{T}{T_0} \right)^4 + \nu_{\text{dyn}} \left[\left(\frac{T}{T_0} \right)^6 + \mathcal{O} \left(\frac{T}{T_0} \right)^8 \right] \quad (2)$$

where $\nu_{\text{sta}} = -2.13023(6)$ Hz [43], $\nu_{\text{dyn}} = -148.7(7)$ mHz [42] and $T_0 = 300$ K. The uncertainty σ_{BBR} of BBR shift includes $\sigma_{\text{BBR},T}$, $\sigma_{\text{BBR},\text{sta}}$ and $\sigma_{\text{BBR},\text{dyn}}$, which are derived from errors of T , ν_{sta} and ν_{dyn} , respectively.

The key to calibrating the BBR shift is to calibrate the temperature T_A at the atom cloud, which can be obtained from the thermal radiation model of $T_A^4 = \sum_j F_{A \rightarrow j} T_j^4$ with the exchange factor $F_{A \rightarrow j}$ and the measured temperature

T_j on the spherical surface j . The uncertainty is limited by $\sigma_{\text{RTDs}} = \sqrt{\sum_j F_{A \rightarrow j}^2} \sigma_{\text{RTD}}$, where σ_{RTD} is the calibration accuracy of one resistance temperature detector. More sensors are helpful to reduce σ_{RTDs} . The inner surface of the 316L stainless steel chamber is polished, and there are 17 sensors (Lakeshore, PT-111) evenly placed on its outer surface, as shown in figure 7(a). Each sensor has a calibration accuracy of 24 mK at 300 K and is traceable to the ITS-90 temperature standard. After considering the acquisition error of the temperature monitor, the overall calibration error σ_{RTD} is 39 mK. Moreover, the corresponding exchange factors are calculated from the ray-tracing analysis based on the emissivity and specular reflectivity values.

To maintain the temperature uniformity and stability, a three-stage temperature control is implemented. The primary air conditioner can maintain the temperature jitter of the laboratory below ± 1 K, and the secondary air conditioner can maintain the temperature jitter of the closed experimental platform below ± 100 mK. Furthermore, the temperatures of the CF35 viewports are actively controlled, allowing the maximum temperature difference between the 17 probes to be less than ± 50 mK within 216 000 s, as shown in figure 7(b). Sufficient measurement time helps reduce the statistical error $\sigma_{T_A, \text{data}}$ of the data. Therefore, the stability of the BBR shift calculated from T_A reaches 1×10^{-19} .

In addition, we built a congruent chamber with an in-vacuum probe T_{in} to verify the correctness of the thermal radiation model. We observe a stability of 6 mK for the difference δT between the in-vacuum probe T_{in} and the thermal radiation model T_A in 90 000 s, even though the environment has a temperature fluctuation range of ± 250 mK, as shown in figure 7(c). The Allan deviation can be used as the statistical error $\sigma_{\delta T, \text{data}}$ of the data. Therefore, the actual temperature at the atom cloud can be corrected to $T = T_A + \delta T$. The total error can be calculated as $\sigma_T = \sqrt{\sigma_{\delta T, \text{data}}^2 + \sigma_{\text{RTD}}^2 + \sigma_{T_A, \text{data}}^2 + \sigma_{\text{RTDs}}^2}$, which is mainly limited by the precision of the in-vacuum

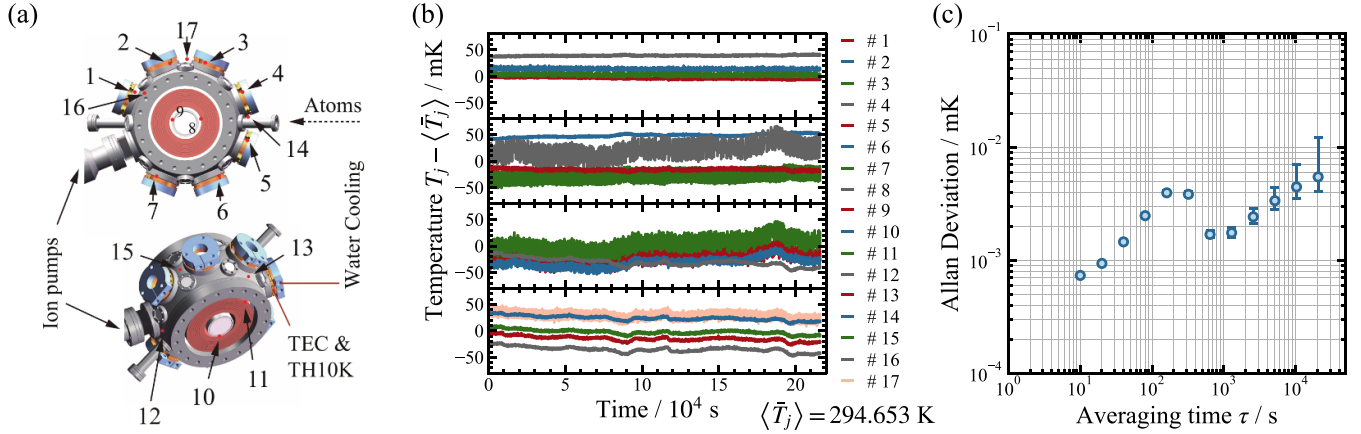


Figure 7. (a) Temperature control and monitor. There are 17 temperature sensors installed on the surface of the main chamber. The temperatures of the CF35 viewports are collected by the probes (Thorlabs, TH10K) and stabilized by thermoelectric coolers (TECs) and water cooling. (b) The long-term temperature fluctuation after active temperature control. $\langle T_j \rangle$ is the mean value of all sensors. (c) The Allan deviation of the temperature offset between the in-vacuum probe and the T_A calculated from the thermal model.

Table 2. The uncertainty of the BBR shift.

Items	Value
σ_{RTD}	39 mK
σ_{RTDs}	11 mK
$\sigma_{T_A, \text{data}}$	<0.1 mK
$\sigma_{\delta T, \text{data}}$	6 mK
σ_T	41 mK
T	294.629 K
$\sigma_{\text{BBR}, T}$	2.8×10^{-18}
$\sigma_{\text{BBR}, \text{sta}}$	0.1×10^{-18}
$\sigma_{\text{BBR}, \text{dyn}}$	1.5×10^{-18}
$\Delta \nu_{\text{BBR}}$	-4926.1×10^{-18}
σ_{BBR}	3.2×10^{-18}

probe (σ_{RTD}). More accurate calibration of the in-vacuum probe will be helpful to reduce uncertainty.

As shown in table 2, T is calibrated as 294.629 K with an uncertainty of 41 mK. After a dynamic coefficient correction about higher order terms, the BBR shift is evaluated as -4926.1×10^{-18} with an uncertainty of 3.2×10^{-18} .

7.2. Zeeman shift

The presence of the biased and ambient magnetic field induces the Zeeman shift. By analyzing the Zeeman splitting difference S_n between $|m_F = \pm 9/2\rangle$ states, the residual 1st-order Zeeman shift can be expressed as

$$\Delta \nu_{B,1} \leq \langle S_{2n} - S_{2n-1} \rangle / 8, \quad (3)$$

and the 2nd-order Zeeman shift can be expressed as

$$\Delta \nu_{B,2} = \beta S^2 \quad (4)$$

with β of $-2.456(3) \times 10^{-7} \text{ Hz}^{-1}$ [42]. We have applied two layers of magnetic shielding to suppress the amplitude

of 50 Hz noise to less than $\pm 0.5 \text{ mG}$. By recording the Zeeman splitting difference, the residual first-order Zeeman shift is evaluated as 3.7×10^{-20} with an uncertainty of 0.8×10^{-18} , and the second-order Zeeman shift is evaluated as -178.5×10^{-18} with an uncertainty of 0.2×10^{-18} .

7.3. Density shift

The density shift comes from cold collisions between atoms, including p-wave and s-wave scattering. In our system, the density shift caused by s-wave scattering is negligible by controlling the polarization imperfection below 2%. However, the density shift coefficient of p-wave scattering is relevant to the lattice trap depth U . The density shift coefficient at a fixed U (dimensionless ratio of trap depth to recoil energy $E_r = \hbar^2 k^2 / 2m$, where k is the amplitude of lattice wave vector, and m is the mass of ^{87}Sr) is calibrated by measuring the frequency shift difference of different atom numbers N . By calibrating these coefficients at lattice traps of $35.1 E_r$, $50.4 E_r$, $76.0 E_r$, $101.2 E_r$, $126.9 E_r$, and $152.0 E_r$, we observe that the density shift is $\propto NU^{3/2}$ and the factor is $-1.030(17) \times 10^{-22} / \text{atom}$ in our system, as shown in figure 8(a). Therefore, the uncertainty of the density shift is derived from the errors of the factors N and U . As shown in table 3, the density shift at the trap depth of $50 E_r$ with 1000 atoms is evaluated as -36.4×10^{-18} with an uncertainty of 0.9×10^{-18} .

7.4. Lattice AC Stark shift

Lattice confinement induces a differential AC Stark shift between the ground and excited clock states. To minimize this effect, the lattice frequency must be set near the magic frequency. Recent works have demonstrated a simplified thermal modeling method to describe non-linear shifts. In our experiment, the differential lattice shifts at these different modulated trap depths of near $35 E_r$, $50 E_r$, $77 E_r$, $100 E_r$, $128 E_r$, and $152 E_r$ relative to the reference depth

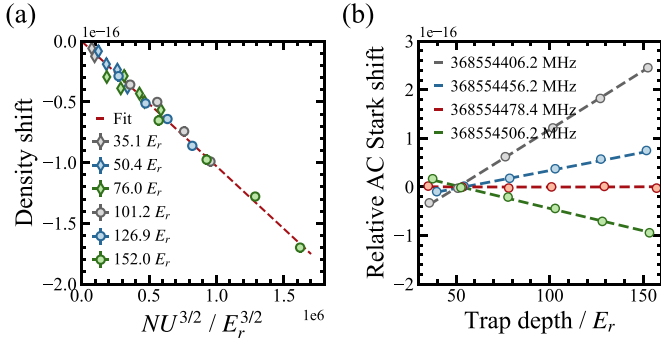


Figure 8. (a) The density shift is fitted with $\propto NU^{3/2}$. (b) Lattice Stark shifts measured related to a trap depth of 50 E_r . The four curves with solid lines are dependently fit using the model in equation (5).

Table 3. The uncertainty of the density shift.

Items	Value	Error	Uncertainty/ 10^{-18}
Factor/ $10^{-22} \text{ atom}^{-1}$	-1.030	0.017	0.6
N/atoms	1000	10	0.4
U	50.0	0.5	0.5
Shift/ 10^{-18}	-36.4		0.9

Table 4. The uncertainty of the lattice AC Stark shift.

Items	Value	Error	Uncertainty/ 10^{-18}
$\frac{\partial \alpha^*}{\partial \nu_1}/10^{-21} \text{ MHz}^{-1}$	33.3	0.2	—
$\beta^*/10^{-22}$	-1.6	2.9	0.7
$\nu_{\text{zero}}/\text{MHz}$	368 554 477.5	1.6	2.7
U	50.0	0.5	<0.1
Shift/ 10^{-18}	0.4		2.8

of 50 E_r are measured at different lattice frequencies ν_1 of 368 554 406.2 MHz, 368 554 456.2 MHz, 368 554 506.2 MHz, and 368 554 478.4 MHz, as shown in figure 8(b). Based on the thermal model method [2, 44], the differential shifts are fitted by the following expression

$$\frac{\Delta \nu_{\text{AC}}}{\nu_{\text{Sr}}} = -\frac{\partial \alpha^*}{\partial \nu_1} (\nu_1 - \nu_{\text{zero}}) U - \beta^* U^2 \quad (5)$$

where α^* and β^* are experiment-specific coefficients and ν_{zero} is the effective magic frequency which is also experimentally determined. Based on the global fit, we obtain the linear coefficient $\frac{\partial \alpha^*}{\partial \nu_1} = 33.3(2) \times 10^{-21} \text{ MHz}^{-1}$, the quadratic term coefficient $\beta^* = -1.6(29) \times 10^{-22}$, and our magic wavelength $\nu_{\text{zero}} = 368 554 477.5 \text{ MHz}$. Therefore, the uncertainty of density shift is derived from errors of U , β^* and ν_{zero} , respectively. The error of $\frac{\partial \alpha^*}{\partial \nu_1}$ has no contribution when $\nu_1 = \nu_{\text{zero}}$. As shown in table 4, the lattice AC Stark shift at the trap depth of 50 E_r with the magic lattice frequency is evaluated as 0.4×10^{-18} with an uncertainty of 2.8×10^{-18} .

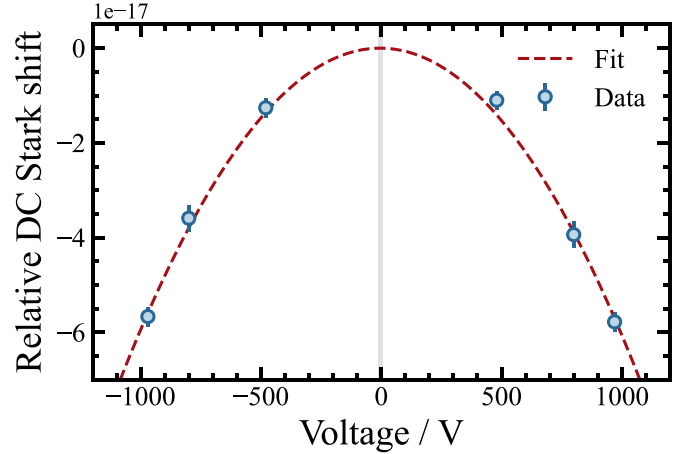


Figure 9. DC Stark shifts measured at different applied voltages V_a relative to the GND. The curve is fit by $\alpha_{\text{DC}}(V_a + V_b)^2 - \alpha_{\text{DC}} V_b^2$, where V_b is the equivalent residual voltage. The gray vertical line represents the position of $-V_b$.

75. DC Stark shift

Stray DC electric fields or patch charges on vacuum viewports can induce a frequency shift due to the differential DC polarizability (α_0) between the two clock states. The DC Stark shift could be modulated by applying different external voltages on a pair of plates fixed on the front and back of the vacuum chamber (along the x -axis in figure 1(a)). $\Delta \nu_{\pm} = -\frac{1}{2} \alpha_0 (E_a \pm E_b)^2$ and $\Delta \nu_{\text{GND}} = -\frac{1}{2} \alpha_0 E_b^2$ are the shifts corresponding to $\pm V$ and the ground (GND), where E_a and E_b correspond to the applied electric field and the background electric field, respectively. The measured experimental quantities are differential shifts between modulated voltages and GND, i.e. $\Delta \nu_{\pm} - \Delta \nu_{\text{GND}}$. Figure 9 shows the measured DC Stark shifts at different applied voltages of $\pm 480 \text{ V}$, $\pm 800 \text{ V}$, and $\pm 970 \text{ V}$ related to GND. Based on the fitting of a quadratic form, V_b is evaluated as $4.8 \pm 9.1 \text{ V}$, which corresponds to a DC Stark shift of 1.4×10^{-21} with an uncertainty of 5.2×10^{-21} . Considering the point charge model, the generated electric potential and electric field strength are proportional to r^{-1} and r^{-2} respectively, resulting in the DC Stark shift proportional to r^{-4} , where r is the distance from the charge on the viewport. As to the other two orthogonal directions, the same amount of patch charges brings a shift with nearly an order of magnitude smaller, due to $r_{y,z}/r_x = 1.67$, where $r_{y,z}$ or r_x is the distance of cavity center from the viewport along the y -axis (z -axis) or x -axis. In addition, the grounded vacuum cavity has a certain shielding effect on the charges. Using finite element analysis, the shielding effect on the shift along the y -axis (z -axis) is three times stronger than the x -axis. Even considering that there are four pairs of viewports, the DC Stark shift along the y -axis (z -axis) does not exceed the x -axis.

76. Background gas collision shift

Two-body collisions between a cold ^{87}Sr atom and a room-temperature hydrogen molecule may cause a systematic shift. The ratio of the background gas collision shift to the atomic

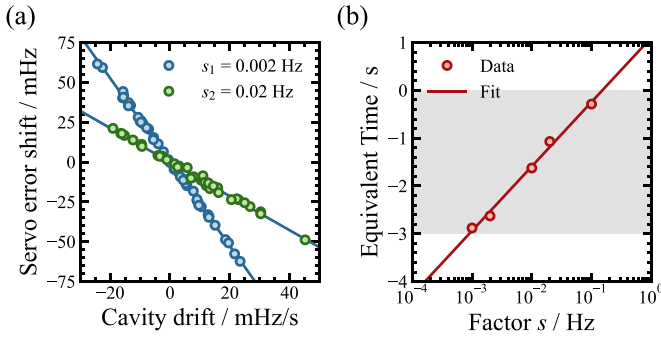


Figure 10. (a) The servo error shifts measured with U1 cavity drift at different interrogation factors. Both the blue and green lines are linear fits. (b) The linear fitting coefficient (corresponding to equivalent time T_s) measured at the different integration parameters s . T_s changes linearly with $\log s$. The shaded area represents the boundary of T_s from $-2T_c$ to 0.

collision loss rate is $-3.0(3) \times 10^{-17}$ [45]. The atomic collision loss rate is determined by the atomic lifetime in lattices 20.5 ± 0.5 s. As a result, the background gas collision shift is evaluated as -1.5×10^{-18} with an uncertainty of 0.2×10^{-18} .

7.7. Servo error shift

The servo error shift can be calculated by postprocessing the excitation fraction. We tested the shift proportional to the U1 cavity drift, as shown in figure 10(a). Moreover, the scale factor can be reduced by increasing the integration factor s of the digital PID algorithm. The cavity drift can be linearly compensated to approximately 0 by the AOM frequency with an interval of 10 ms. Figure 10(b) shows that T_s changes linearly with $\log s$, in which the drift can be calibrated to $\pm 0.1 \text{ mHz s}^{-1}$. With a scale factor of $T_s = -0.29$ s in our setup, the uncertainty of the servo error shift is evaluated as 0.7×10^{-19} .

7.8. Other shifts

Benefitting from the long clock pulse of 1390 ms, the probe AC Stark shift, line pulling shift, and lattice tunneling shift are calculated at the 10^{-20} or even 10^{-21} level.

For the AOM phase chirp shift, we repeatedly measured the phase (ϕ) jitter in the FNC link before and after the AOM is turned on. The fluctuation of excitation fraction δp can be analyzed by $\delta p = \frac{1}{2} \int_0^{T_p} \frac{d}{dt} \phi(t) \cdot g(t) dt$. By warming the AOM and reducing the operating power of the RF, the phase jitter is not apparent, resulting in an AOM chirp shift of -0.3×10^{-21} with an uncertainty of 2.7×10^{-21} .

Based on the above evaluation results, the uncertainty of our Sr1 optical clock is 4.4×10^{-18} .

8. Conclusion

In this article, we present the physical system of the ^{87}Sr optical lattice clock, in which the ^{87}Sr atom cloud is laser-cooled and loaded into a 1D optical lattice with a lifetime of

more than 20 s and a temperature of a few micro-Kelvin. An ultrastable laser system has been built to generate the clock beam for interrogating atoms. We obtain narrow clock carrier spectra with a linewidth of sub-Hertz, which allows us to adopt an interrogation time of more than 1 s. We have achieved the closed-loop operation of our optical clock and measured the cross-spectrum density of the clock laser to analyze the Dick-effect-limited instability. Then, we implement the evaluation of instability by interleaved measurement under different conditions, from which the stability of the optical lattice clock can be improved to $4.5 \times 10^{-16}/\sqrt{\tau}$ by increasing the clock pulse and decreasing the dead time, which is consistent with the theoretical calculation. Furthermore, the stability of a single clock is evaluated by frequency comparison between Sr1 and Sr2 clocks. From that, we have analyzed the systematic instability of $5.4 \times 10^{-16}/\sqrt{\tau}$ and the instability of $3.0 \times 10^{-16}/\sqrt{\tau}$ from the physical noise. Moreover, the total instability of the individual clock reaches 2.1×10^{-18} at an averaging time of 47 000 s. In addition, we evaluated the uncertainty of different systematic frequency shifts, resulting in a low total uncertainty of 4.4×10^{-18} . In the future, we will implement an optical clock frequency comparison between different atom species, such as Sr/Yb [46] or Sr/Ca⁺ [47, 48].

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ORCID iDs

Xiao Jiang <https://orcid.org/0000-0002-1026-5075>
 Han-Ning Dai <https://orcid.org/0000-0003-0916-7942>
 Yu-Ao Chen <https://orcid.org/0000-0002-2309-2281>
 Jian-Wei Pan <https://orcid.org/0000-0002-6100-5142>

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